



AIAA 3111 - Introduction to Data Mining

Course Information and Basic Introduction

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AIoT · Edge Intelligence · Smart Mobility

Research Interest

- Artificial Internet of Things
- Human-Centric Sensing & Computing
- Mobile & Edge Intelligence
- Smart Mobility & Urban Intelligence

Academic Outcomes/Services

- **PI of three NSFC grants** (including 2 general programs)
- More than **60 papers** published in top journals and conferences, including ACM MobiCom, MobiSys, KDD, IEEE ICDE, ICDM and IEEE TKDE, IEEE TMC, IEEE TITS, along with **14 field patents**
- **Associate Editor of IEEE TETCI**
- **IEEE/CCF Senior Member**



WeChat

From Sensing to Intelligence: Building Intelligent Systems for the Physical World

Selected Research Projects

#1 Human-Centric Sensing & Intelligence

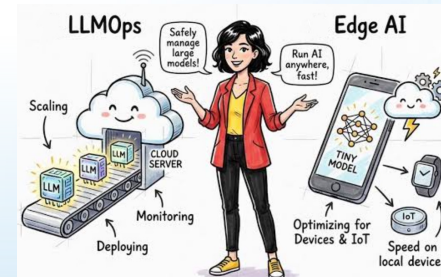


#2 Smart Mobility & Urban Intelligence



- Traffic Sensing (TMC'24, TITS'26)
- Traffic Prediction (TITS'25)
- GPS Enhancement (TMC'24)
- Ride-Sharing (ICDE'20, IOTJ'22, TMC'24,)
- Transportation Dispatching (TITS'22, TMC'25)
- Data Synthetic (DASFAA'21, TMC'26)
- Mobility Analysis (TMC'23, TKDE'25, CIKM'25-26)
- Traffic Signal Control - Astra (KDD'26)

#3 Edge AI & Intelligent Systems



- (1) On-device/edge deployment → inference
- (2) Sense → Learn → Reason → Act

(ACM MobiCom, SenSys, UbiComp, IEEE TMC)

Source: <https://www.instagram.com/p/DXpBrG8IXRO/>

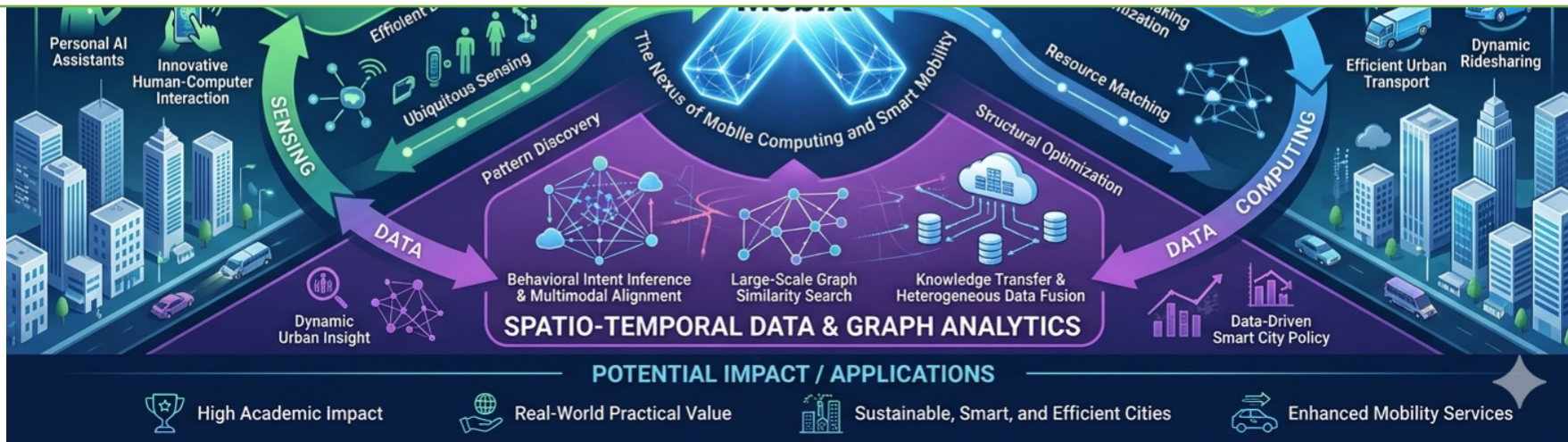
About Zhidan LIU's Research Group

MobiX Research Group

The **MobiX** (*Mobile Intelligence & eXploration*) research group is dedicated to advancing the vision of smart mobility. We explore and develop innovative systems driven by efficient mobile sensing, ubiquitous computing, and mobility data analytics. Our research spans across urban computing, the Artificial Internet of Things (AIoT), and spatio-temporal data mining. The mission of **MobiX** is to conduct cutting-edge research in these intersecting fields, delivering robust solutions that yield both high academic impact and real-world practical value.



Welcome UG Students to join us, to explore new research problems via X-Program or competitions or other collaborations.



Logistics

Time: **Monday 15:00-16:20PM / Friday 10:30-11:50AM**

Venue: **Lecture Hall A, in person**

Instructor	Email	Office	Office Hour
Zhidan Liu	zhidanliu@hkustgz.edu.cn	W4-306	Tuesday 15:00-17:00

Teaching Assistant:

Name	Email	Thrust
WANG, Jiyu	jwang738@connect.hkust-gz.edu.cn	INTR
LI, Cong	cli166@connect.hkust-gz.edu.cn	DSA



Intended Learning Outcomes (ILOs)

1. Demonstrate **thorough knowledge** of the *literature* and a comprehensive understanding of *scientific methods* and techniques relevant to *data mining*.
2. Demonstrate **practical skills** in building *data mining* systems.
3. Critically **apply** theories, methodologies, and knowledge to address *fundamental questions* in *data mining*.
4. Independently **pursue research or innovation** of significance in *data mining* applications.



Course Schedule

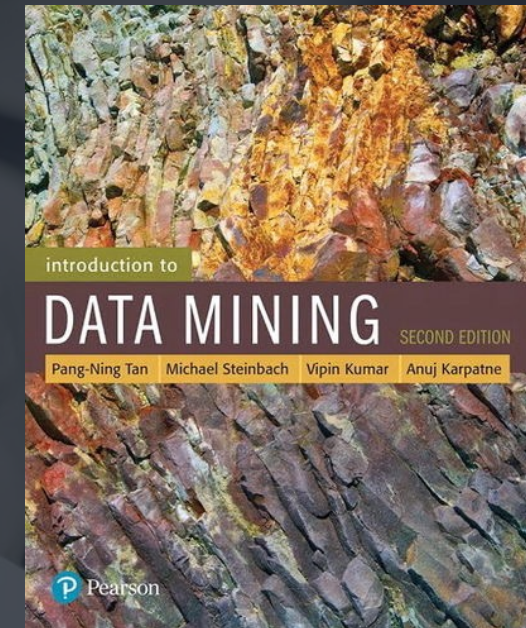
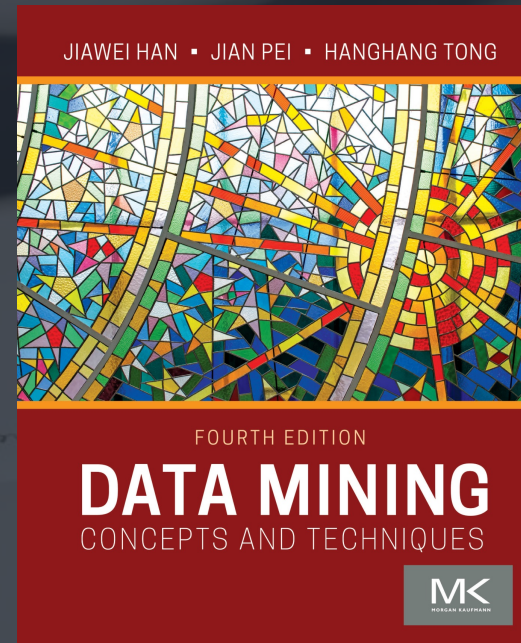
Week #	Topic
1	Introduction
2	Data Preprocessing
3	Pattern Mining I
4	Pattern Mining II
5	Classification (Basic)
6	Classification (Advanced)
7	<i>Midterm exam</i>
8	Clustering I
9	Clustering II
10	Anomaly Detection
11	Data Mining on Diverse Data
12	Advanced LLMs and Agents in Data Mining
13	Group Poster Presentation and Q&A
TBD	<i>Final exam</i>



References

Books:

- [Data Mining: Concepts and Techniques, 4th ed.](#), by Jiawei Han, Jian Pei and Hanghang Tong
- [Introduction to Data Mining \(Second Edition\)](#), by Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar



Grading

Final grade will be composed of

- Course participation: **10%**
- Writing Assignment: **20%** (four assignments)
- Mid-term exam: **25%**
- Course project: **20%**
 - Poster presentation in the last week
- Final exam: **25%**



Grading

Group Project & Poster

- **Group Formation:** 2-4 members, forming your group ASAP.
- **Poster:** A poster presentation (5-minute intro. and Q&A), scheduled for one week before the final exam.
- **Project Focus:** Application of Data Mining to real-world problems.
- **Project Topic:**
 - Option 1: Form a team and brainstorm issues or problems of personal interest;
 - Option 2: If needed, we will provide a list of potential project topics to help guide your exploration.
This list will be released as soon as possible, but no later than the midterm exam.
- **Grading Rubrics:** The grading rubrics for the group project and poster presentation will be released after the midterm exam.



Honor Code

- We strictly enforce the HKUST Honor Code

Make sure you read and understand it!

The HKUST Academic Honor Code

Honesty and integrity are central to the academic work of HKUST. Students of the University must observe and uphold the highest standards of academic integrity and honesty in all the work they do throughout their program of study.



As members of the University community, students have the responsibility to help maintain the academic reputation of HKUST in its academic endeavors.



Sanctions will be imposed on students, if they are found to have violated the regulations governing academic integrity and honesty.

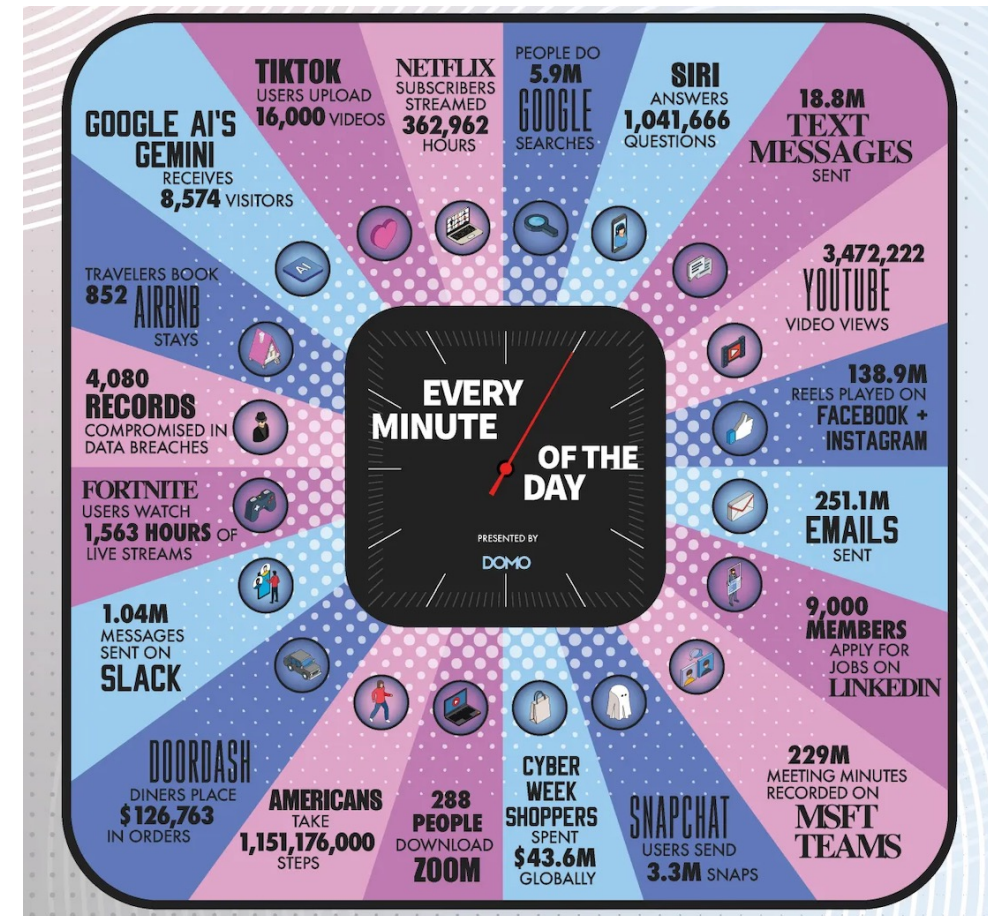
Tips on Proper Use of Generative AI

- ChatGPT and related LLMs are revolution tools and are encouraged to use in your study and research, i.e., light editing, idea exploration
- Direct use the content generated by ChatGPT and other LLMs is prohibited

Learn to use the tool to improve yourself rather than slack off!

Large-scale Data is Everywhere!

- ❑ We live in a world where *vast amounts of data* are generated *constantly* and *rapidly*
 - ❑ Transaction data in stores, credit cards
 - ❑ Scientific measurements
 - ❑ Health records, brain images, daily measurements
 - ❑ Queries to Google, Clicks, Browsing behavior, Ads
 - ❑ Social network data, e.g., Meta, X, Weibo
 - ❑ Movement data, Trajectories
 - ❑ Mobile use, telephone calls
 - ❑ Wearable devices
 - ❑ Machine and workflow monitoring



Data Creates Value

LLMs (like ChatGPT, DeepSeek) relies on massive amounts of data for training

DA

I'm looking to write marketing copy to make my email marketing more engaging. The product I'm promoting is an office cleaning service. What subject lines would you suggest for a sequence of 3 emails. I want to send 3 welcome emails for new prospects.



Here are a few suggestions for subject lines for a sequence of 3 welcome emails for a new prospect promoting an office cleaning service:  

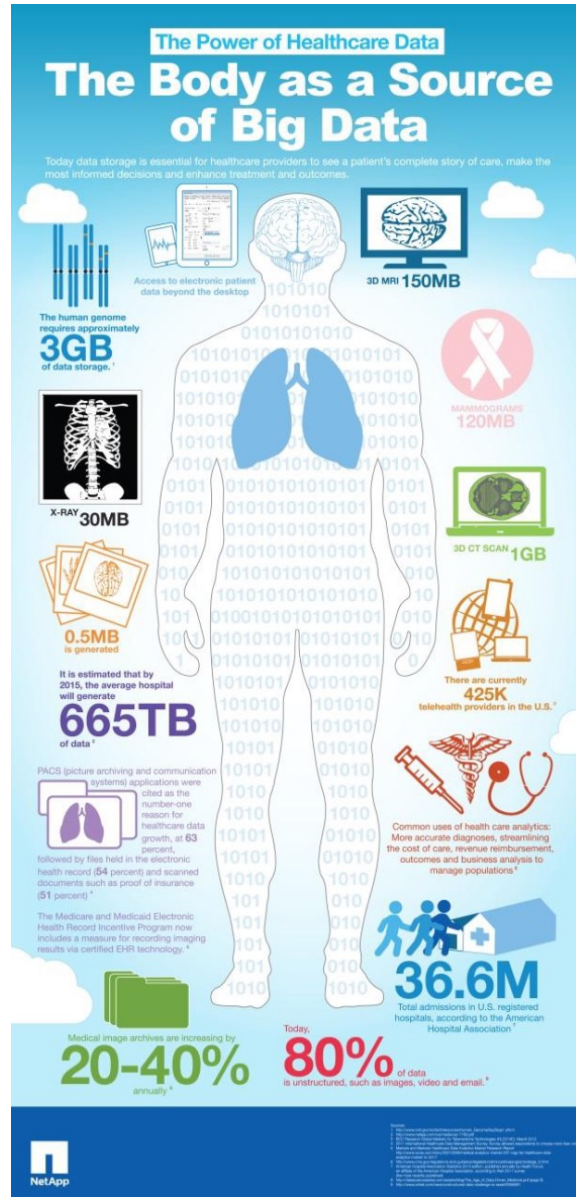
1. "Say Goodbye to Messy Offices with Our Top-Notch Cleaning Services"
2. "Improve Your Employee's Wellness and Productivity with Our Cleaning Services"
3. "Upgrade Your Office's Appearance with Our Professional Cleaning Services"

You may also want to consider including a sense of urgency or scarcity in your subject lines to encourage the recipient to open and read the email right away. For example, "Limited Time Offer: Get 15% off Your First Cleaning Service" or "Don't Miss Out on Our Exclusive Cleaning Package Deals."

It's also important to keep in mind that the subject line should be relevant to the content in the email and should accurately reflect what the recipient can expect when they open the email.

Another tip is to use personalization in the subject line by using recipient's name, company name, or location which will increase the open rate as it will look more personalized and less like spam.

Data Creates Value



Precision/Personalized medicine:

Find the best treatment for patients using their genotype and all data related to them

Also: understanding drug side-effects through massive data records

Data Creates Value

The self-driving car's sensors

Just like a person has five senses, Google's self-driving car has a variety of gadgets that detect nearby objects so it can avoid them.

Global Positioning System software
Helps car determine its location.

Position sensor
Located in the wheel hub, this sensor helps determine car's location from wheel rotations.

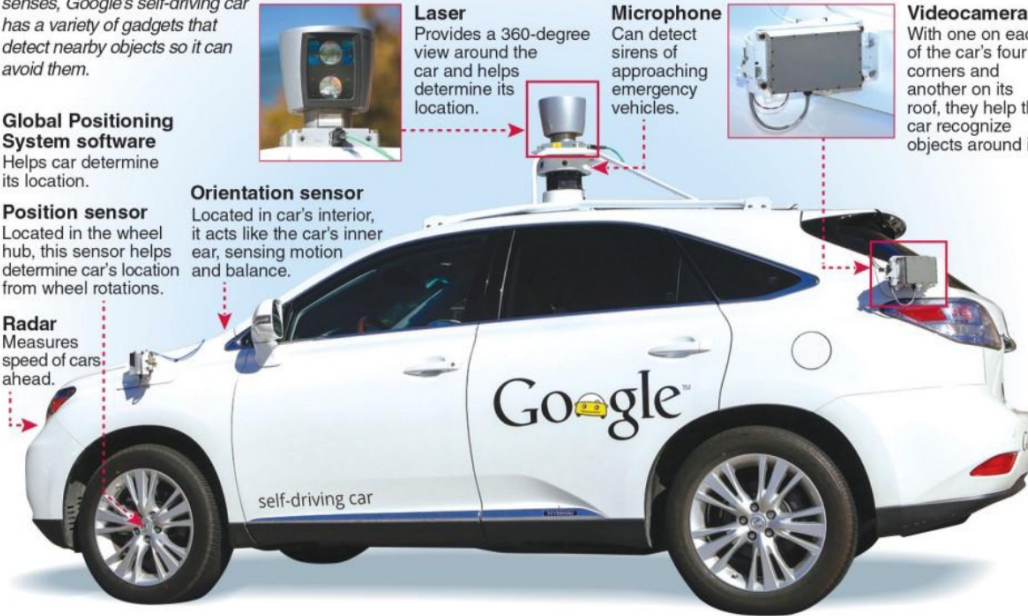
Radar
Measures speed of cars ahead.

Orientation sensor
Located in car's interior, it acts like the car's inner ear, sensing motion and balance.

Laser
Provides a 360-degree view around the car and helps determine its location.

Microphone
Can detect sirens of approaching emergency vehicles.

Videocameras
With one on each of the car's four corners and another on its roof, they help the car recognize objects around it.



Self-Driving Cars:

Car is the next computer. A future of smart cars that can drive themselves and learn from data

Also: smart cities – urban computing

How the car operates

- 1 Any object the vehicle's sensors spot is interpreted by software to determine if it's a pedestrian, cyclist, vehicle or something else.
- 2 Using what it's learned from previous driving, the software makes predictions about what objects will do next.
- 3 The software analyzes the information to decide whether it is safe to accelerate, turn or hit the brakes.



How the car sees the world

This computerized image is what Google researchers monitoring sensor data see as they ride in the vehicle.

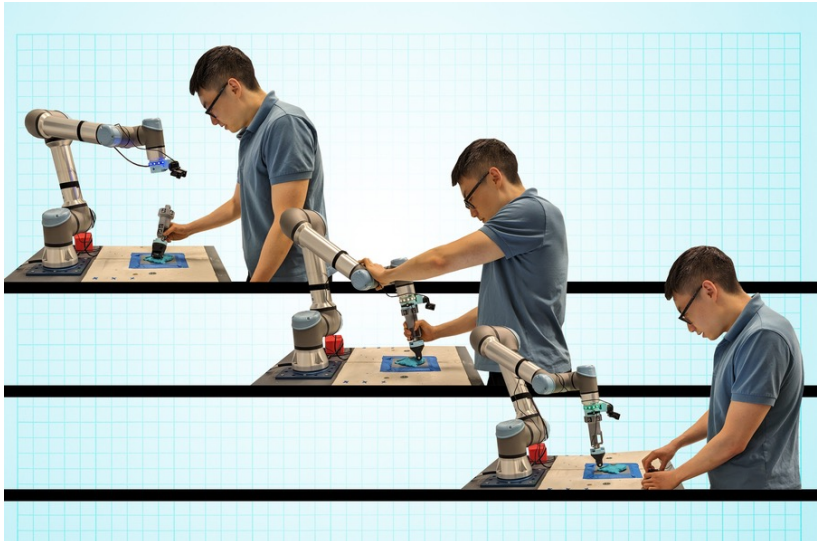
- Other vehicle
- Pedestrian
- Cyclist
- Objects that warrant caution
- A crosswalk, indicating the car needs to stop
- A traffic signal, warning of upcoming railroad tracks
- Path where Google's car intends to go

Source: Google
Graphic: Tribune News Service

Data Creates Value

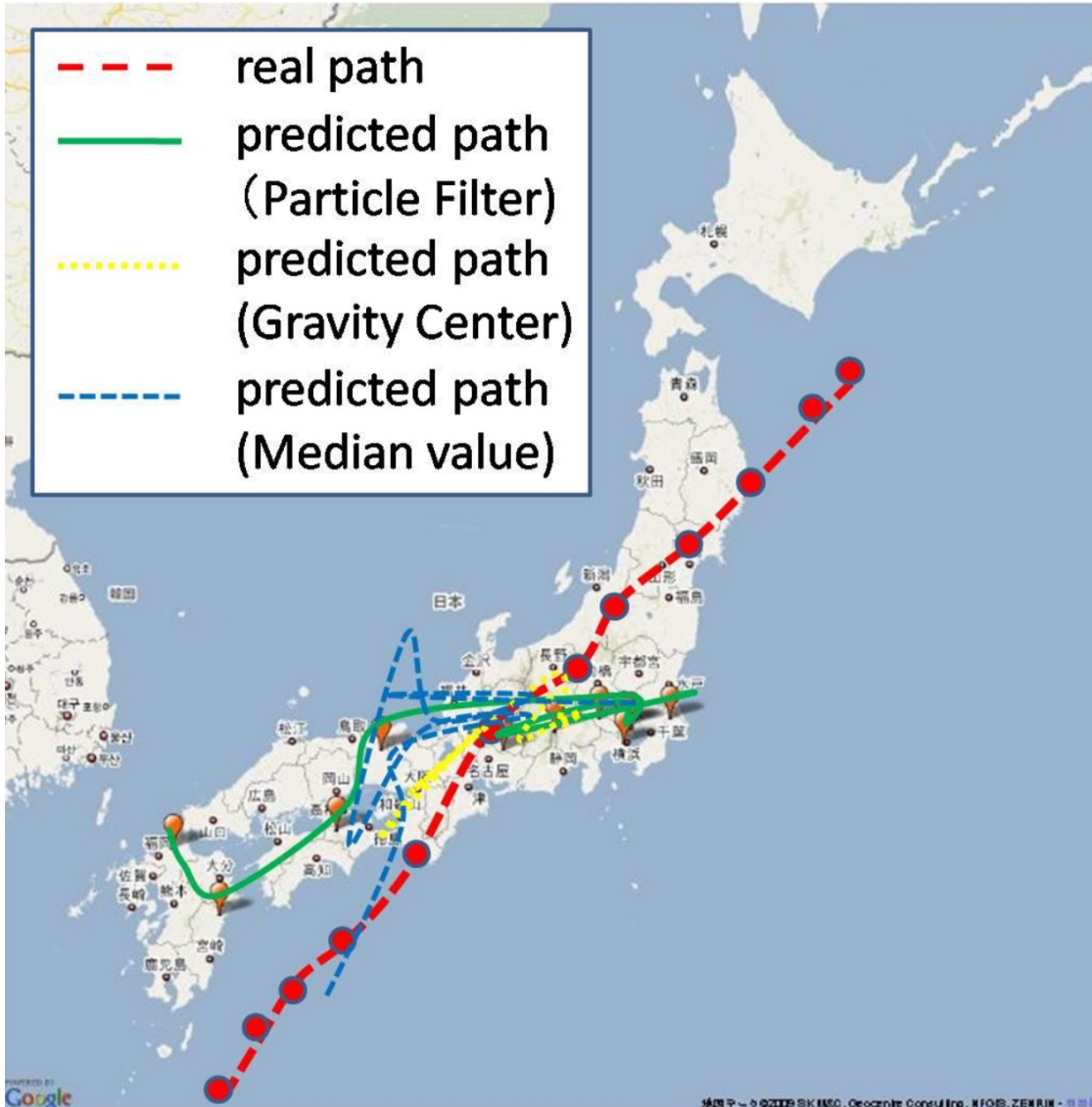


Computers learn to play games by observing data



Robots can learn from demonstrations

Data Creates Value



Use of data for crisis management

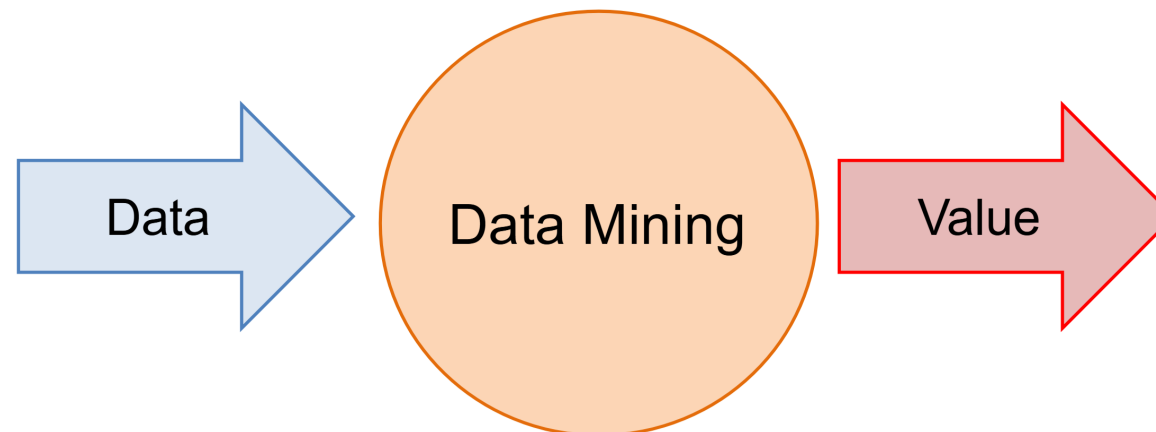
The data is **complex** and **interconnected**

- ❑ Multiple types of data: tables, text, time-series, images, videos, graphs, etc.
 - ❑ Spatial and temporal aspects
 - ❑ Multimodal data
- ❑ Interconnected data of different types:
 - ❑ From the mobile phones, we can collect location of the user, friendship information, check-ins to venues, opinions through twitter, status updates in WeChat, images through cameras, queries to search engines

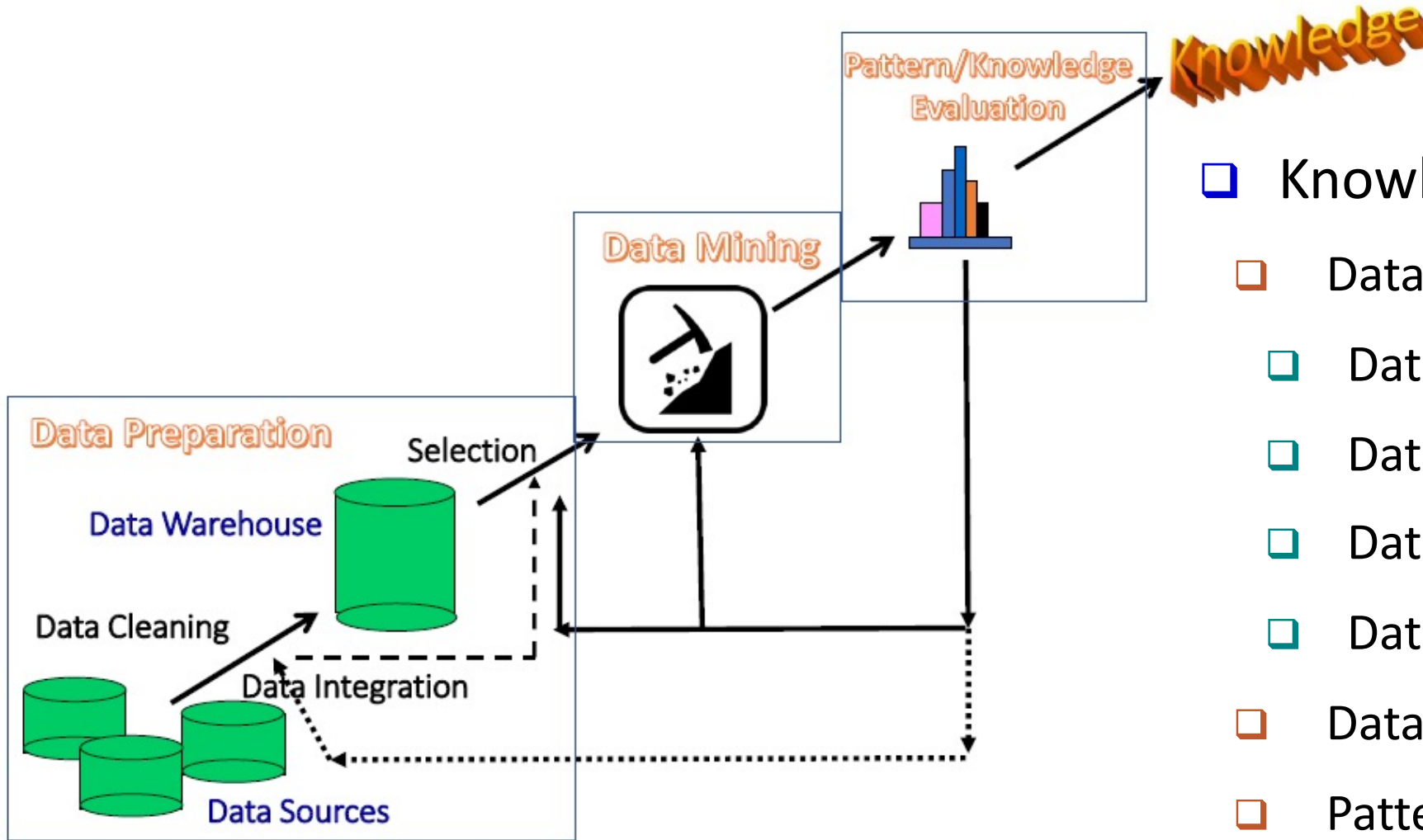


What Is Data Mining?

- ❑ **Data mining** is the process of discovering interesting patterns, models and other kinds of knowledge in large data sets
 - ❑ “Data mining”: a misnomer? It should be “**knowledge mining from data**”
 - ❑ Other terms: *Knowledge mining from data*, **KDD** (*Knowledge Discovery from Data*), *pattern discovery*, *knowledge extraction*, *data analytics*, *information harvesting*
- ❑ “Data Mining is the study of **collecting, processing, analyzing**, and **gaining useful insights** from data” – Charu Aggarwal (@IBM)



Data Mining: An Essential Step in Knowledge Discovery



Knowledge Discovery Process

- Data preparation
 - Data cleaning
 - Data integration
 - Data transformation
 - Data selection
- Data mining
- Pattern/model evaluation
- Knowledge presentation

Data Types for Data Mining

Structured vs. unstructured data

- Structured: uniform, record- or table-like structures, defined by data dictionaries, with a fixed set of attributes, each with a fixed set of value ranges and semantic meaning
- Eg. Data stored in *relational databases*, *data cubes*, *data matrices*, and many *data warehouses*

	A	B	C	D	E	F	G
1	Purchase ID	Last name	First name	Birthday	Country	Date of purchase	Amount of purchase
2	1	Davidson	Michael	04/03/1986	United States	10/12/2016	37
3	2	Vito	Jim	09/01/1994	United Kingdom	02/02/2016	85
4	3	Johnson	Tom	23/08/1972	France	02/11/2016	83
5	4	Lewis	Peter	18/10/1979	Germany	22/11/2016	27
6	5	Koenig	Edward	13/05/1983	Argentina	26/03/2015	43
7	6	Preston	Jack	16/06/1991	United States	06/11/2016	77
8	7	Smith	David	11/03/1965	Canada	15/11/2016	23
9	8	Brown	Luis	03/09/1997	Australia	03/07/2015	74
10	9	Miller	Thomas	07/01/1980	Germany	07/11/2016	13
11	10	Williams	Bill	26/07/1960	United States	20/11/2015	80
12	11	Gemini	Alexia	12/09/1995	Canada	11/03/2017	35
13	12	Bond	James	25/02/1975	United Kingdom	12/08/2017	40
14	13	Burgle	Patricia	01/12/1990	United States	18/01/2015	55
15	14	Reding	Michelle	07/04/1985	Canada	23/02/2017	28
16	15	Harvey	Billy	14/07/1971	United Kingdom	12/01/2016	41
17							

Structured Data (Tabular Data)

Data Types for Data Mining

❑ Structured vs. unstructured data

- ❑ *Semi-structured*: allow a data object to contain a set value, a small set of heterogeneous typed values, or nested structures, or to allow the structure of objects or sub-objects to be defined flexibly and dynamically
- ❑ Data having *certain structures* with **clearly defined semantic meaning**, such as *transactional data set*, *sequence data set* (e.g., time-series data, gene or protein data, or Weblog data)

```
{
  "patient_id": "P003",
  "timeline": [
    {"t": "2025-08-30T08:55:00Z", "event": "Arrival-ED"},
    {"t": "2025-08-30T09:02:00Z", "event": "ECG", "result": "ST-elevation?"},
    {"t": "2025-08-30T09:20:00Z", "event": "Lab-Troponin", "value": 0.38}
  ]
}
```

sequence data (Semi-Structured)

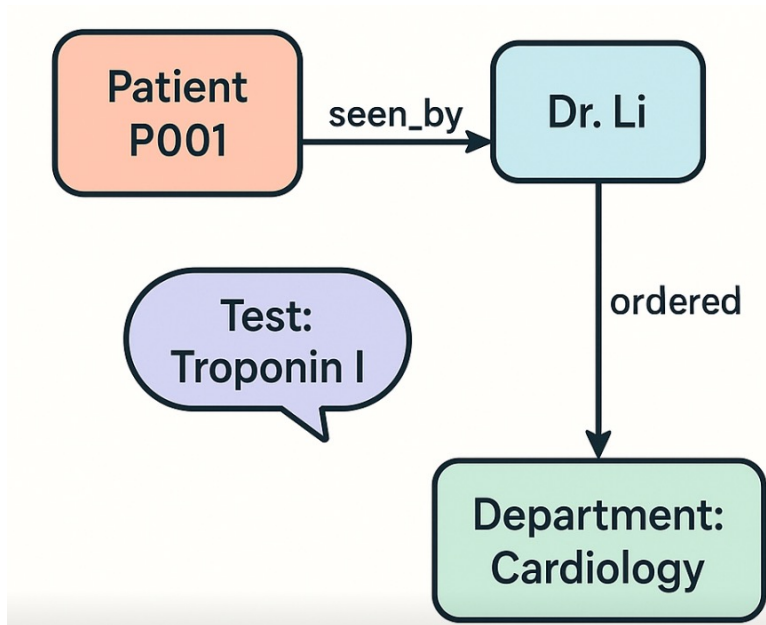
```
{
  "transaction_id": "TX-100231",
  "patient_id": "P001",
  "timestamp": "2025-09-01T12:00:05Z",
  "items": [
    {"drug": "Aspirin", "dose": "100mg", "qty": 30},
    {"drug": "Atorvastatin", "dose": "20mg", "qty": 30}
  ],
  "payment": {"method": "Card", "amount": 28.60}
}
```

transactional data (Semi-Structured)

Data Types for Data Mining

❑ Structured vs. unstructured data

- ❑ *Graph or network data*: A more sophisticated type of **semi-structured** data set
- ❑ *Unstructured data*: text data and multimedia (e.g., audio, image, video) data



Graph or Network

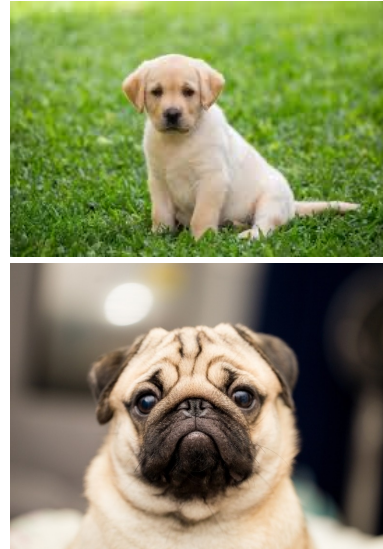
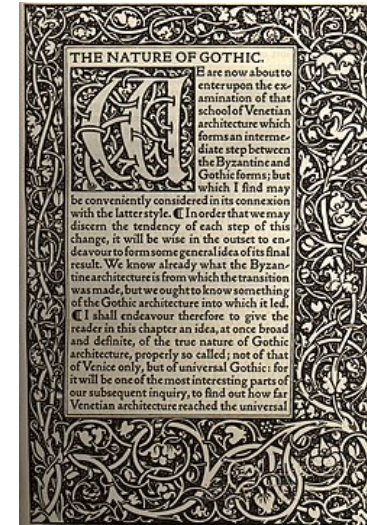


Image (Unstructured Data)

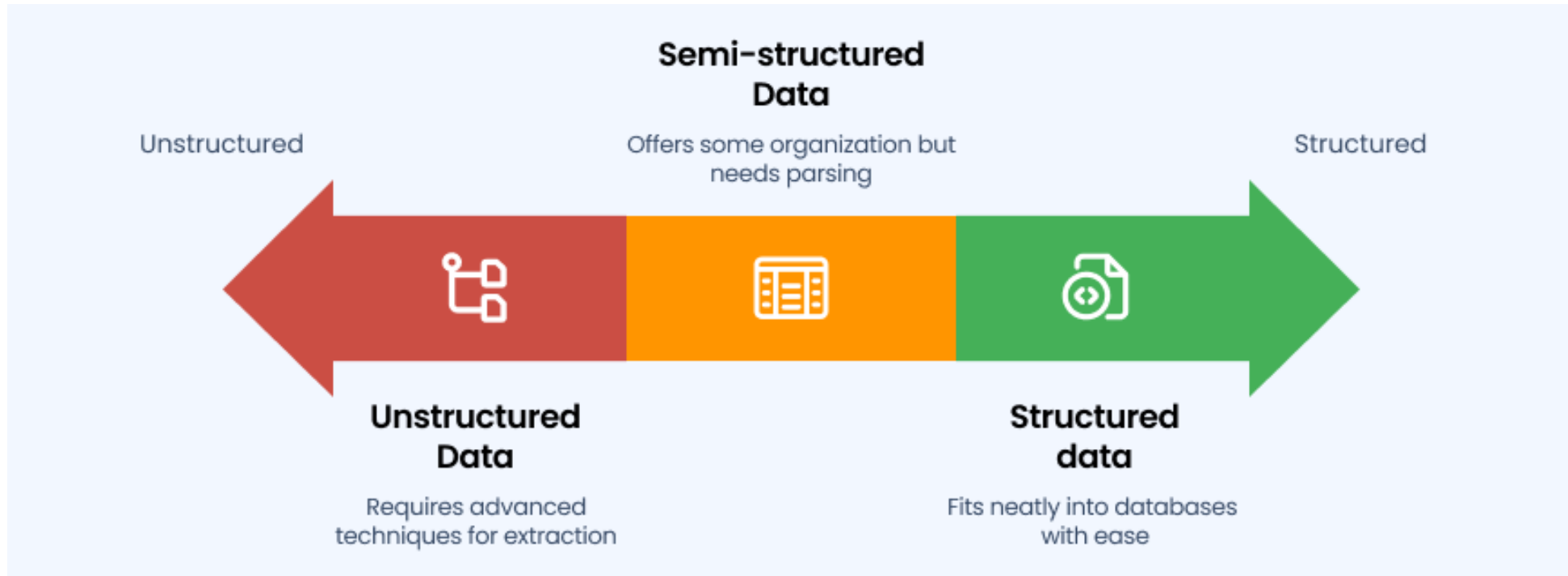


Text (Unstructured Data)

Chief complaint:
chest tightness for
2 hours.
ECG shows ST
elevation in V2–V4.
Recommend
troponin test and
immediate
cardiology consult.

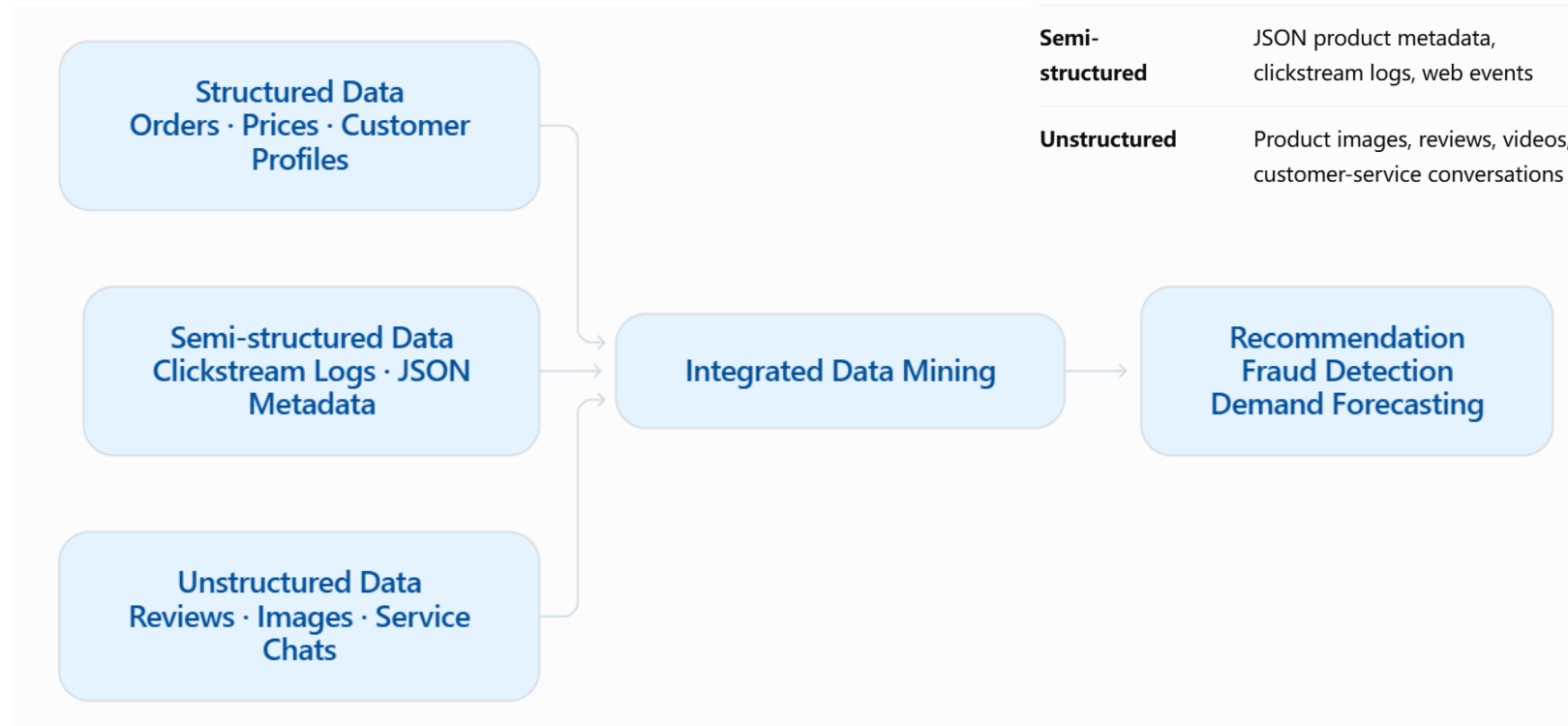
Data Types for Data Mining

□ Structured vs. Semi-structured vs. unstructured data



Real-World Data Is Often Heterogeneous

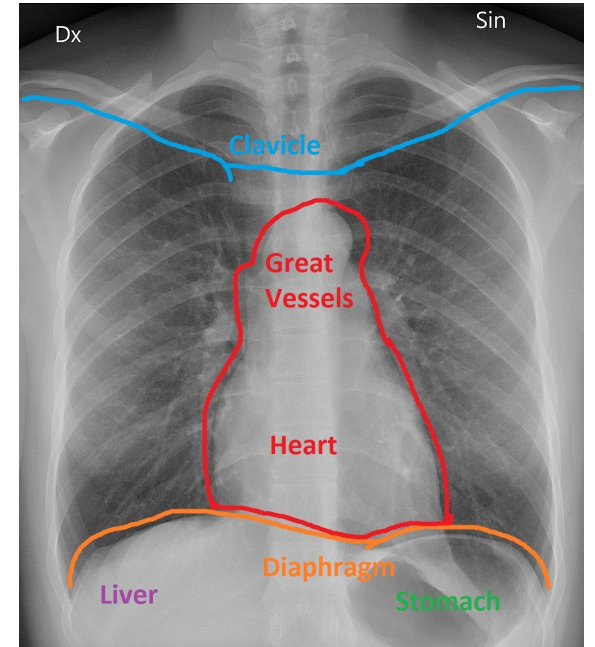
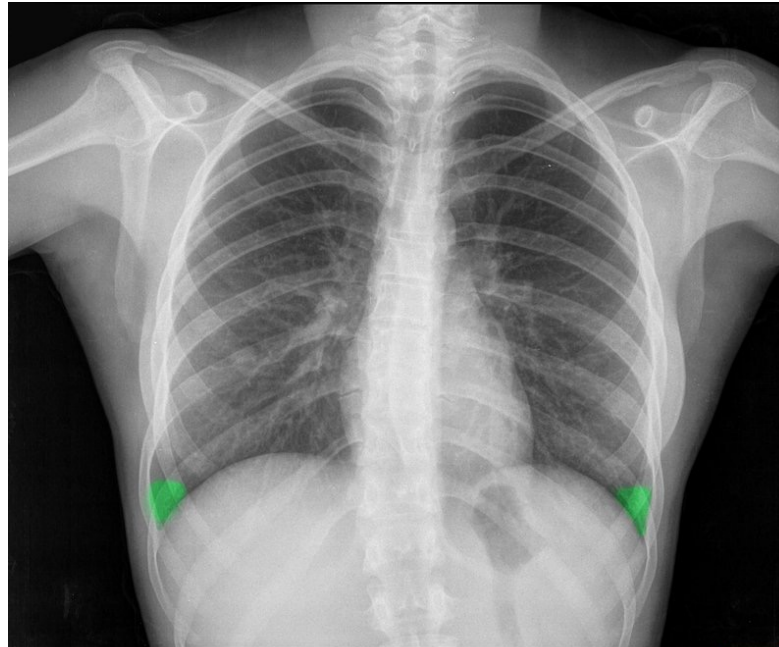
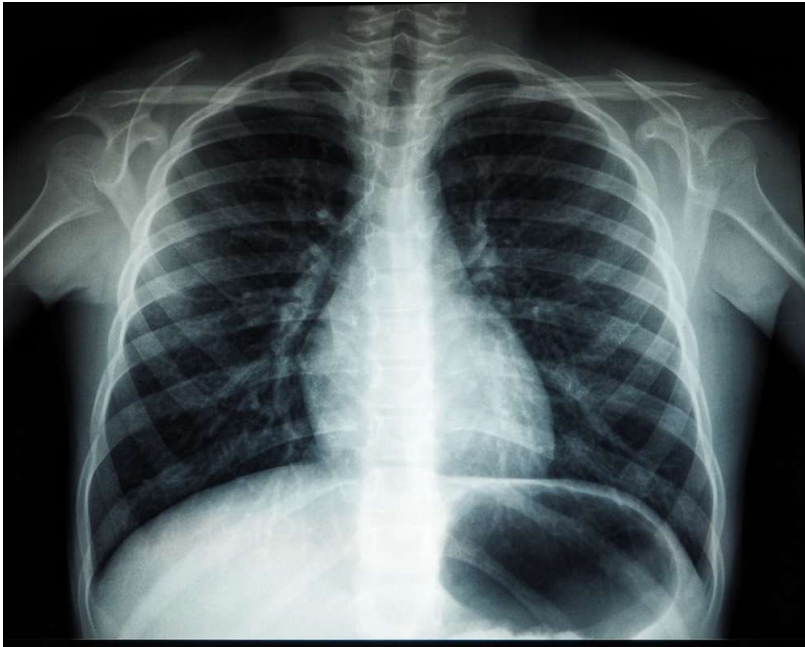
- Real-world data often consists of a mixture of structured, semi-structured, and unstructured data.



Data type	E-commerce examples	Characteristics
Structured	Order tables, product prices, customer profiles	Fixed schema; organized in rows and columns
Semi-structured	JSON product metadata, clickstream logs, web events	Contains tags or keys, but does not follow a rigid table schema
Unstructured	Product images, reviews, videos, customer-service conversations	No predefined data model; requires feature extraction or representation learning

Data Types for Data Mining

Question: Semi-Structured or Unstructured Data?



Data Type Alone Does Not Determine the Mining Method

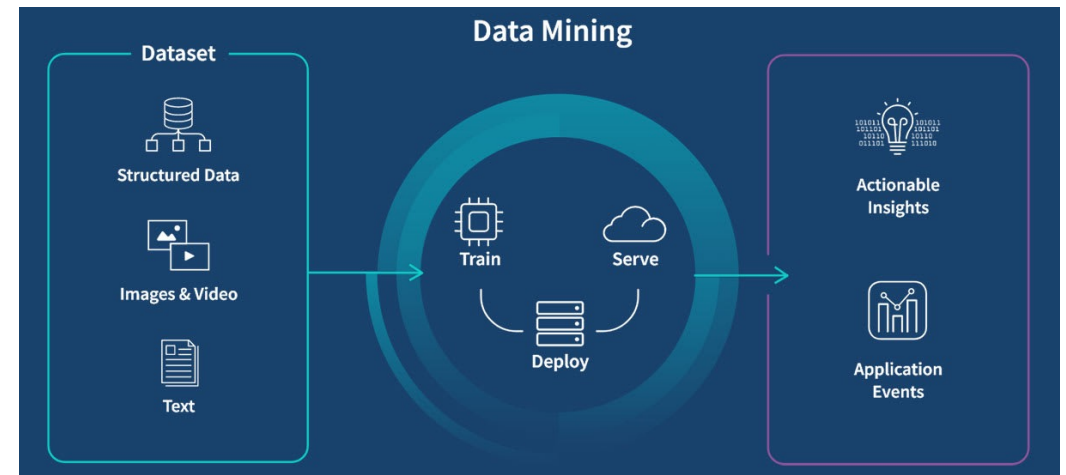
Same Structure, Different Semantics

- ❑ Both are sequences, but their meanings differ.
 - ❑ DNA sequence: discover motifs and mutations
 - ❑ Purchase sequence: discover preferences and next-item behavior

Same Data, Different Questions

- ❑ Mobility Records
 - ❑ Predict future demand → Prediction
 - ❑ Group similar regions → Clustering
 - ❑ Find unusual trips → Anomaly Detection
 - ❑ Discover common routes → Pattern Mining

Different semantics and analytical goals require different mining methods.



How Data Arrives Changes How It Is Mined

❑ Stored vs. streaming data

- ❑ Stored data: Finite, stored in various kinds of large data repositories
- ❑ Streaming data (e.g., video surveillance or remote sensing): Dynamic, constantly coming, infinite, real-time response—posing challenges on effective data mining

	Batch / Stored Data	Streaming Data
Scope	Finite and stored	Continuous and potentially unbounded
Access	Can be processed repeatedly	Usually processed in one or a few passes
Priority	Accuracy and comprehensive analysis	Low latency and limited memory
Examples	Historical transactions, archived records	Sensor feeds, network traffic, live clickstreams
Typical methods	Offline and batch algorithms	Online and incremental algorithms

For streaming data, decisions must often be made before the complete dataset is available.

What Can We Discover from Data?

❑ Data Summarization

Describe the main characteristics of a dataset

❑ Frequent Pattern and Associations Mining

Discover items or events that frequently occur together

❑ Classification and Regression

Predict categorical labels or numerical values

❑ Cluster Analysis

Discover natural groups without predefined labels

❑ Outlier Analysis

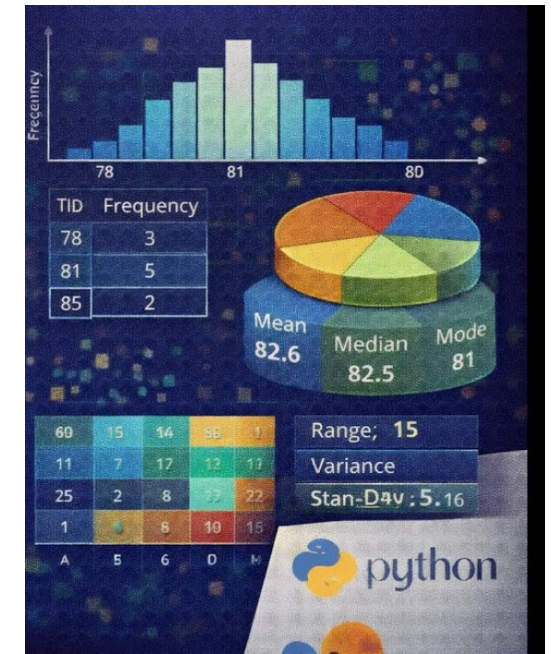
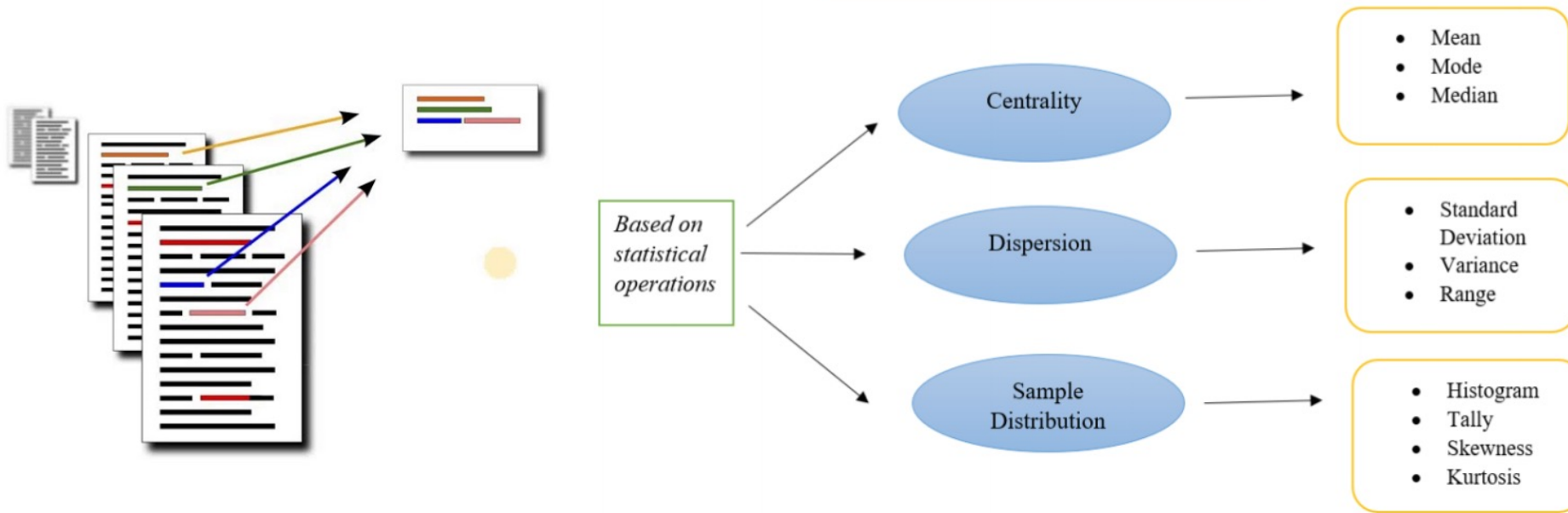
Identify unusual or unexpected observations

Five Core Questions in Data Mining

Question	Task
What is typical?	Data Summarization
What occurs together?	Pattern Mining
What will happen?	Classification & Regression
What groups naturally exist?	Clustering
What is unusual?	Anomaly Detection

Data Summarization: What Is Typical?

- ❑ **Goal:** Provide a compact and interpretable description of a dataset
- ❑ **Example:** Summarize sales by product, region, and month
- ❑ **Outputs:** Aggregates, profiles, comparisons, and visual summaries



Pattern Mining: What Occurs Together?

- ❑ **Input:** A collection of transactions or event sequences
- ❑ **Goal:** Discover frequently co-occurring items or events
- ❑ **Example:**
 - ❑ {Smartphone} → {Phone Case}
 - ❑ Diaper → Beer [0.5%, 75%] (support, confidence)
- ❑ **Key measures:** Support, confidence, and lift

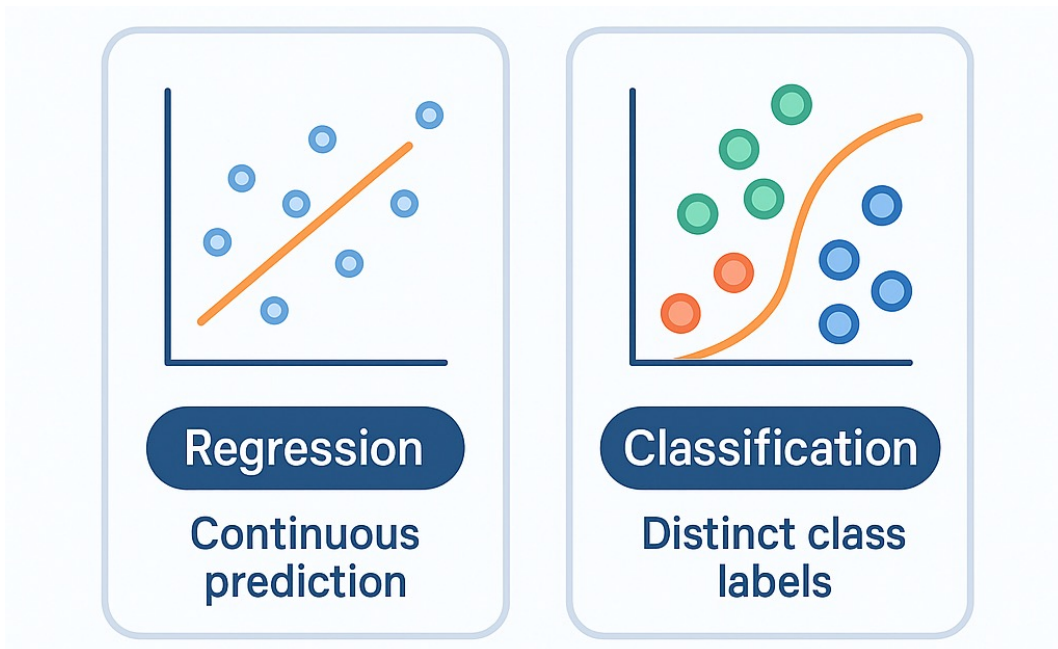


$$\begin{aligned} \text{Rule } X \Rightarrow Y & \begin{cases} \text{Support} = \frac{\text{Frequency}(X,Y)}{N} \\ \text{Confidence} = \frac{\text{Frequency}(X,Y)}{\text{Frequency}(X)} \\ \text{Lift} = \frac{\text{Support}}{\text{Support}(X) * \text{Support}(Y)} \end{cases} \end{aligned}$$

Association does not necessarily imply causation.

Prediction: What Will Happen?

- ☐ Predictions: **Classification** or **Regression**
 - ☐ Construct models (functions) based on some training examples (with labels)
 - ☐ Describe and distinguish classes or concepts for future prediction



Classification

Predict a categorical label

Spam or not spam?

Fraud or legitimate?

Regression

Predict a numerical value

How long will a trip take?

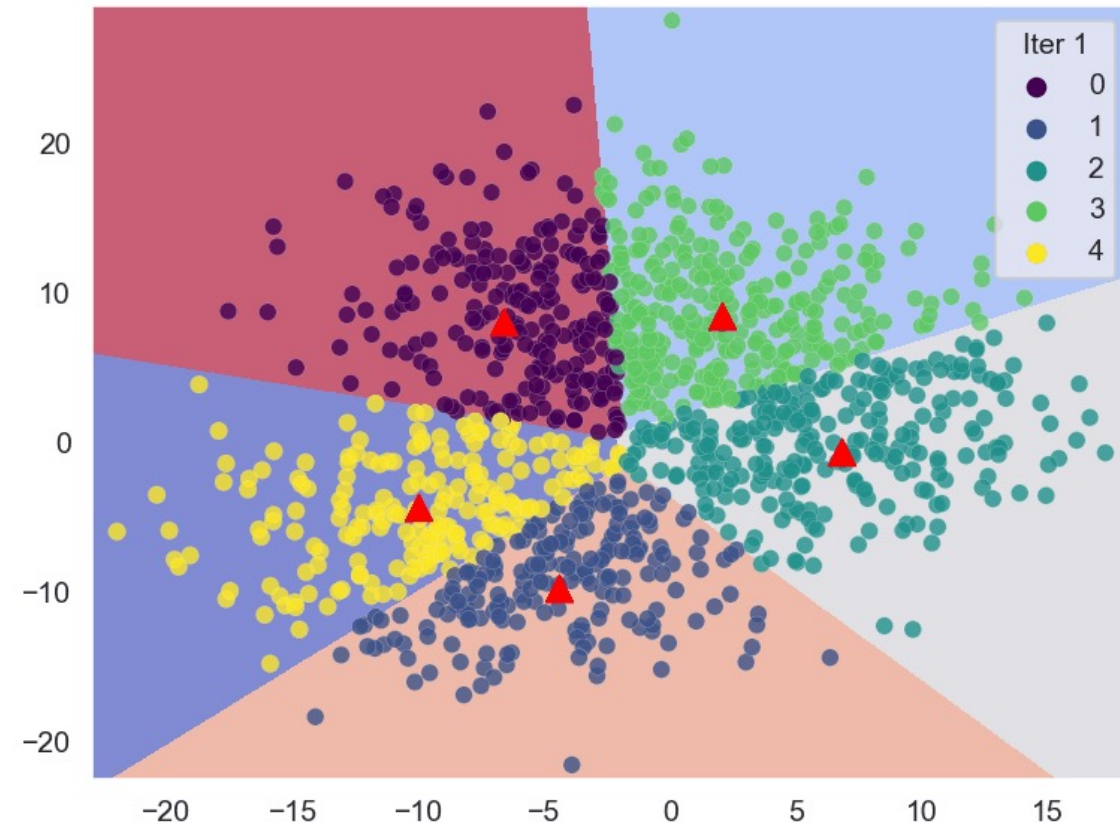
How much demand will occur?

Both learn predictive relationships from labeled examples.

Clustering: What Groups Naturally Exist?

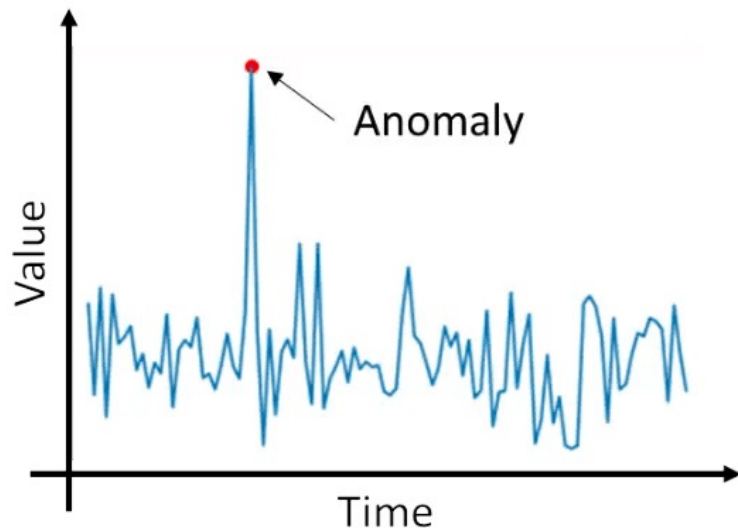
- ❑ **Input:** Unlabeled data (Unsupervised learning)
- ❑ **Goal:** Discover groups of similar objects
- ❑ **Principle:** High similarity within clusters and low similarity between clusters
 - ❑ Maximize within-cluster similarity and minimize between-cluster similarity.
- ❑ **Example:** Group urban regions by their mobility patterns

Classification predicts known classes;
clustering discovers unknown groups.



Anomaly Detection: What Is Unusual?

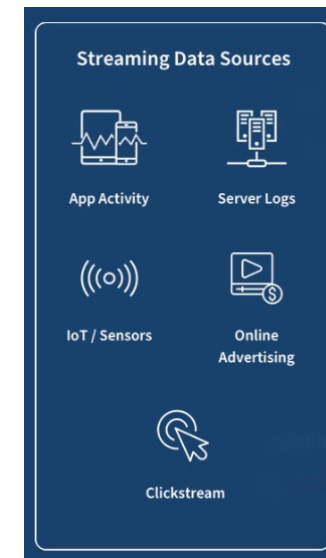
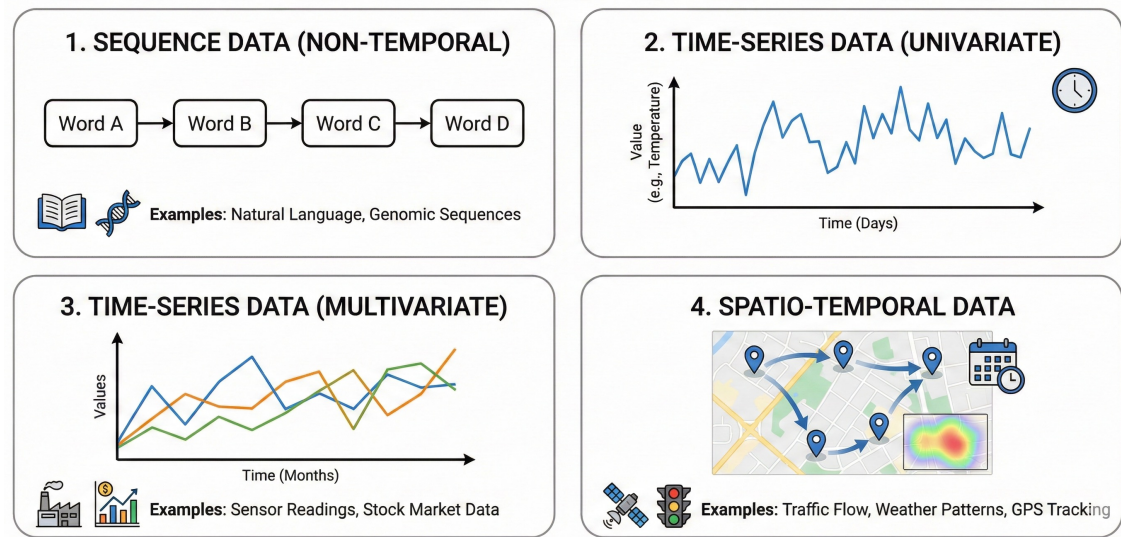
- ❑ Outlier analysis
 - ❑ Outlier: A data object that does not comply with the general behavior of the data
 - ❑ It may represent an **error, rare event, or new phenomenon**
 - ❑ Whether something is anomalous often depends on its **context**
 - ❑ Noise or exception?—One person's garbage could be another person's treasure



An outlier may be noise - or the most valuable observation.

Mining Ordered and Evolving Data

- Sequence, time-series, and evolution analysis
- Mining data streams: Ordered, time-varying, potentially infinite, data streams



Data	Question	Example
Sequence	What events frequently follow one another?	Phone → Case → Earbuds
Time series	How does a value change over time?	Traffic demand
Data stream	What is happening right now?	Live sensor readings

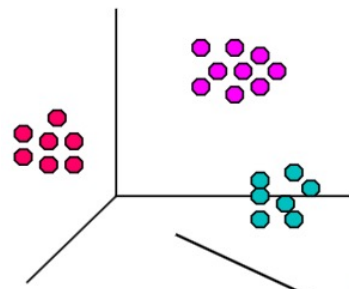
Order and time-series introduce *trends*, *dependencies*, and *changes* that ordinary tables may not capture.

Tasks and Techniques Are Different

Data Mining Tasks—What?	Techniques—How?
Classification	Decision trees
Regression	Statistical models
Clustering	K-means
Pattern mining	Apriori
Anomaly detection	Isolation-based methods
Multiple tasks	Neural networks / Deep learning

Deep learning is a family of techniques that can support multiple data mining tasks - not a separate mining task.

Example: Data Mining Tasks & Techniques

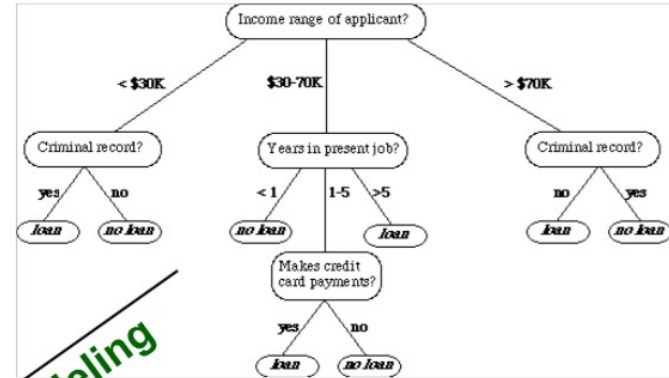


Clustering

Data

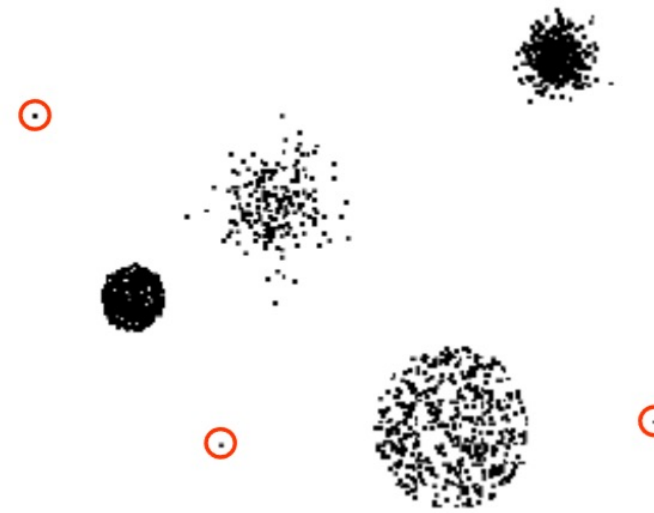
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes
11	No	Married	60K	No
12	Yes	Divorced	220K	No
13	No	Single	85K	Yes
14	No	Married	75K	No
15	No	Single	90K	Yes

Association Rules



Predictive Modeling

Anomaly Detection



Not Every Discovered Pattern Is Useful

- ❑ Data mining algorithms may discover thousands of candidate patterns.
Many are trivial, redundant, unstable, or irrelevant to the user.
- ❑ How do we identify interesting patterns?
 - ❑ **Validity**: Does the pattern hold on new or unseen data?
 - ❑ **Novelty**: Does it reveal something unexpected or previously unknown?
 - ❑ **Utility**: Can it support a meaningful decision or action?
 - ❑ **Interpretability**: Can users understand and explain the pattern?

Objective Measures	Subjective Measures
Accuracy	Novelty
Support / coverage	Usefulness
Statistical significance	Actionability
Robustness on new data	Interpretability

Interestingness depends on both
statistical evidence and **user needs**.

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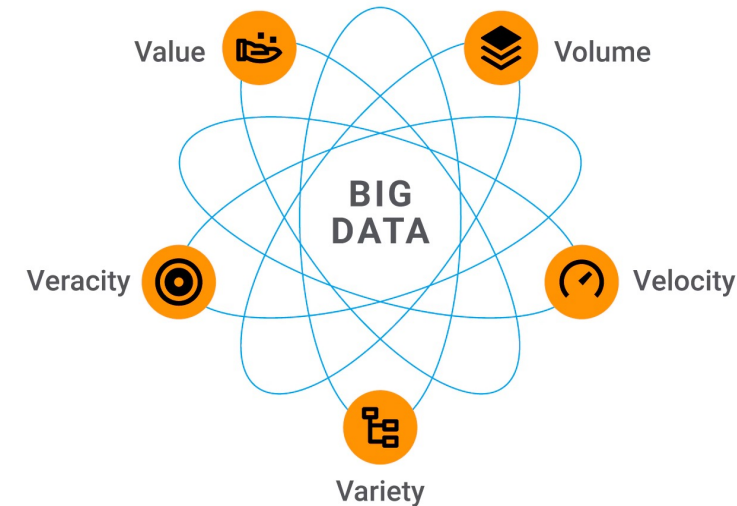
- Example: *“Traffic is heavier during rush hours.”*

- Valid? Novel? Useful?
- Probably / No / It depends

A pattern can be statistically valid but practically uninteresting.

Why Is Real-World Data Mining Challenging?

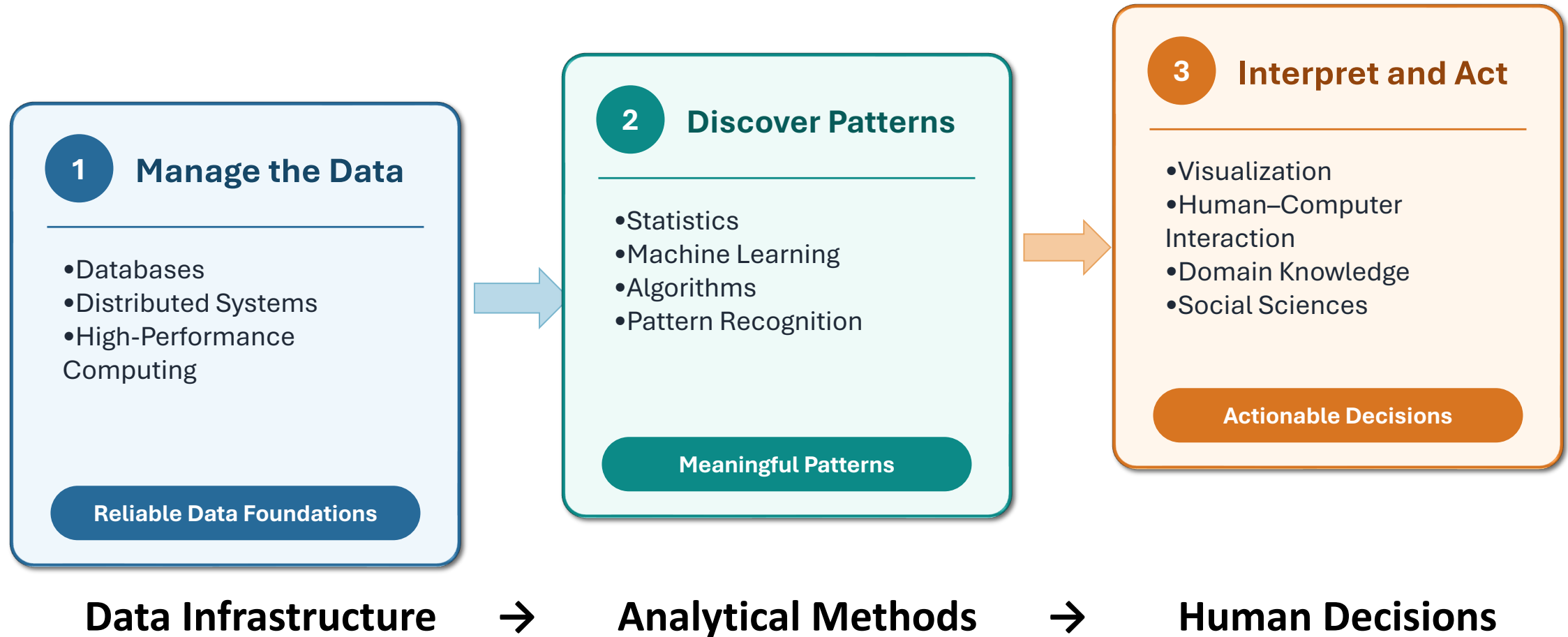
- ❑ **Scale**
Millions or billions of records
- ❑ **High dimensionality**
Thousands or even millions of features
- ❑ **Data heterogeneity**
Tables, text, images, sequences, graphs, and multimodal data
- ❑ **Dynamic environments**
Streaming data and patterns that change over time
- ❑ **Context and user needs**
A statistically valid pattern may not be useful in practice



Real-world data mining requires **scalable algorithms**, **reliable systems**, and **domain understanding**.

Data Mining Brings Multiple Disciplines Together

A progressive path from reliable data foundations to actionable decisions



AI, Machine Learning, Data Mining, and Data Science

- These fields overlap, but they emphasize different questions and stages of the data lifecycle.

Field	Central Goal	Typical Focus
Artificial Intelligence	Build systems that exhibit intelligent behavior	Reasoning, planning, perception, learning
Machine Learning	Learn models and representations from data	Prediction and generalization
Data Mining	Discover valid, novel, and useful patterns from data	Patterns, clusters, anomalies and predictive insights
Data Science	Turn data into evidence and decisions	The end-to-end data lifecycle

Data Mining Is Embedded in Everyday Systems

❑ Digital Platforms

Search ranking, content recommendation, spam detection

❑ Commerce and Finance

Market-basket analysis, demand forecasting, fraud detection

❑ Science and Healthcare

Disease risk analysis, biomarker discovery, scientific data analysis

❑ Cities and IoT

Traffic prediction, mobility analysis, sensor anomaly detection



Data mining often works behind the interface, turning behavioral and sensor data into decisions.

Useful Data Mining Must Also Be Responsible

- ❑ **Privacy and consent**

Was the data collected and used appropriately?

- ❑ **Bias and fairness**

Do the data and models disadvantage particular groups?

- ❑ **Transparency and accountability**

Can decisions be explained, questioned, and corrected?

- ❑ **Security and misuse**

Could the data or discovered patterns be abused?

Can we do it? → Should we do it? → Who may be affected?

Summary

❑ Five Takeaways from Today

- ❑ Data appears in many **forms and representations**
- ❑ Different questions define different **mining tasks**
- ❑ Discovered patterns must be **evaluated**
- ❑ Data mining combines **algorithms, systems, and domain knowledge**
- ❑ Useful mining must also be **responsible**

